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How to do USB On-the-go (OTG) on USB Type-C?

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🕒 Mar 17, 2017 Knowledge

Article Number

000011173

Title

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Article URL

<https://support.microchip.com/s/article/How-to-do-USB-On-the-go-OTG-on-USB-Type-C> (<https://support.microchip.com/s/article/How-to-do-USB-On-the-go-OTG-on-USB-Type-C>)

Question

How to do USB On-the-go (OTG) on USB Type-C?

Answer

Problem Description:

In the many years prior to the availability of USB Type-C, the USB Micro-A/B interface provided a means of selecting the initial role of Host or Device mode operation of a USB product, by altering the termination of the “ID” pin. Similarly, Host Negotiation Protocol (HNP) is a methodology thru which the Host/Device role can be exchanged while a USB session is already established.

USB Type-C introduces a similar concept for how to establish the initial role of USB Host/Device , using the new CC signal on the interface. The initial Host/Source* is the side of the connection which entered the USB-C Attached state with its “Rp” termination applied to the CC wire, and the initial Device/Sink* is the side of the connection which entered USB-C Attached state with its “Rd” termination applied that CC wire.

Once the port partners are in Attached state, the Rp or Rd terminations cannot be exchanged unless USB Power Delivery protocol is also used upon the USB Type-C connection, and an explicit Power Role Swap request was accepted.

- USB Type-C specification describes default port roles in terms of which side is supplying VBUS first in the connection, so the “Host” role is associated with the default “Source” in the connection, and likewise for Device/Sink. During a USB Type-C connection, the port partner with Rp applied to CC is the VBUS Source, and likewise for Rd/VBUS Sink.

So how can host/device role be swapped upon a USB Type-C connection then?

Solution:

All existing OTG methodologies (ID pin, Host Negotiation Protocol, and Session Request Protocol) are prohibited from being used on USB Type-C ports!

Table 5 of the USB Type-C Product Matrix denotes OTG as “Not compatible with USB Type-C”:



Table 5: USB-C Test Matrix

Product Type	Testing Required						
	USB-C CabCon	USB-C EPC	USB PD	USB-C Functional	USB-C IOP	USB-C Source Power	USB 3.1 and 2.0 ¹
Cable	X	X ²	X ³				
Charger & Battery Pack			X	X	X	X	X ⁴
Host & Hub				X	X	X	X
PD Host & PD Hub			X	X	X	X	X
Alt Mode Only Host			X	X	X		
Device				X	X		X
PD Device			X	X	X		X
OTG	Not compatible with USB Type-C						

Reference: http://www.usb.org/developers/compliance/usbcpd_testing/USB-C_Product_Matrix_2017_01_17.pdf (http://www.usb.org/developers/compliance/usbcpd_testing/USB-C_Product_Matrix_2017_01_17.pdf)

The only permitted method of swapping Host or Device role is through the use of USB Power Delivery protocol, following the Data Role Swap command sequence. This requires a USB Power Delivery controller to achieve and simply changing the Rp/Rd termination during a USB Type-C connection does not achieve the role swap.

For detail of USB Power Delivery protocol, proceed to download the latest USB specification distribution and view the contents of the */USB Power Delivery* folder within.

URL Name

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